



MEDICARE



A Project Report

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Requirements for the award of the
Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

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ABSTRACT

CanBridge: Multi-Cancer Detection and Support System is a web-based healthcare platform that integrates AI-assisted cancer screening with structured emotional and informational support. The system is designed to bridge the gap between early detection and post-diagnosis guidance, addressing both the technical and human dimensions of cancer care.

The platform utilizes deep learning models built using TensorFlow/Keras and deployed through a FastAPI backend to analyze medical images for breast and skin cancer screening. The breast cancer model is trained on the BUSI ultrasound dataset, while the skin cancer model is trained on the HAM10000 dermoscopic dataset. Each model performs binary classification and provides a confidence score to assist in preliminary screening.

The frontend is developed using Streamlit, offering a clean and accessible user interface. Users can upload images for screening, view prediction history, and download structured AI-generated reports. All user data, prediction results, and community interactions are managed using a SQLite database with secure authentication mechanisms.

Beyond detection, CanBridge incorporates support-oriented features including Warrior Stories, categorized discussion forums, financial assistance listings of verified NGOs and government schemes, and educational cancer information pages. These components aim to reduce emotional isolation and improve awareness among patients and families.

The system emphasizes ethical AI deployment, user privacy, and clear medical disclaimers. By combining AI-driven screening with community and financial support features, CanBridge moves beyond conventional detection systems and provides a more comprehensive and humane digital health support ecosystem.

ACKNOWLEDGEMENT

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-TANISHA JOSHI

ARATION

I hereby declare that the project entitled, “**MEDICARE-UNUSED MEDICINCE EXCHANGE SYSTEM**” done at **SVKM’s Usha Pravin Gandhi College of Arts, Science and Commerce**, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)** to be submitted as final semester project as part of our curriculum.

-Tanisha Joshi

TABLE OF CONTENTS

SR.NO	CONTENT		PAGE
			NO
1	Introduction		
	1.1	Background	13
	1.2	Objectives	13
	1.3	Purpose, Scope, and Applicability	14
		1.3.1 Purpose	14
		1.3.2 Scope	15
		1.3.3 Applicability	15
	1.4	Achievements	17
	1.5	Organization of Report	17
2	Survey of Technologies		
	2.1	Frontend	19
		2.1.1 Comparison With Framework	19
		2.1.2 Python	19
		2.1.3 Streamlit	20
		2.1.4 Features Of Frontend Stack	21
		2.1.5 Advantages	21
		2.1.6 Disadvantages	21

		2.1.9	Summarization	22
	2.2	Backend		23
3	Requirement Analysis			
	3.1	Problem Definition		27
	3.2	Requirement Specification		27
		3.2.1	Functional Requirements	27
		3.2.2	Non-Functional Requirements	28
	3.3	Planning and Scheduling		28
		3.3.1	Project Planning Process	28
		3.3.2	Scheduling Process	29
		3.3.3	Techniques used for Scheduling (Gantt chart)	31
	3.4	Software Requirements		33
		3.4.1	Developer Side	33
		3.4.2	Client-Side	34
	3.5	Hardware Requirements		35
		3.5.1	Developer Side	35
		3.5.2	Client-Side	35
	3.6	Preliminary Product Description		37

		Conceptual Models	37
	3.7	3.7.1 SDLC	37
		3.7.2 Agile Model	37
		3.7.3 Why Agile Model ?	38

4	System Design			
	4.1	Basic Modules		40
	4.2	Database Design		42
		4.2.1	User Table Design	42
		4.2.2	Data integrity and constraints	42
	4.3	Procedural design (Logic diagrams)		45
		4.3.1	Use Case	45
		4.3.2	Activity	46
		4.3.3	Sequence	47
		4.3.4	Entity Relationship	48
	4.4	User Interface Design		49
	4.5	Security Issues		51
4.6	Test Case Design		53	

5	Implementation and Testing		
	5.1	Implementation Approaches	54
	5.2	Testing Approach	65
		5.2.1	Testing Methodologies

	5.2.2	Test Cases	66
6	Results and Discussion		
	6.1	Overview	71

	6.2	Landing and About us pages		72
		6.2.1	Landing	72
		6.2.2	About	73
	6.3	Login and Signup pages		74
		6.3.1	Login	74
		6.3.2	Signup	75
	6.4	Dashboard		76
	6.5	Disease Prediction Page		77
	6.6	Medicine Recommendation Page		78
	6.7	Nutrition Plan Page		79
	6.8	Workout Plan Page		80
	6.9	Disease Description Page		81
	6.10.	Additional Interfaces		82
		6.10.1	OCR	82
		6.10.2	ChatBot	83

		6.10.3	Medical Text Analyzer	84
7	Conclusion			
	7.1	Conclusion		85
	7.2	Limitations of the system		86
	7.3	Future Scope of the project		87
References				89

LIST OF TABLES			
SR.NO		TABLE NAME	PAGE NO
2.	2.1.1	Comparisons with frameworks	19
	2.1.9	Summarization	22
	2.2.2 .4	MySQL V/S MongoDB	26
4.	4.2	Database Design	42-43
	4.6	Test Case Design	53
5.	5.2.1	Testing Methodologies	65

LIST OF FIGURES		
SR.NO	FIGURE NAME	PAGE NO

3.	3.1	Gantt Chart	32
	3.2	Agile Model	36
4.	4.1	Use case diagram	45
	4.2	Activity Diagram	46
	4.3	Sequence Diagram	47
	4.4	Entity relationship diagram	48
	4.5	User Interface: CureWise-Landing page	49
	4.6	User Interface: CureWise-Login/signup page	50
6.	6.1	Landing/About us Page	72
	6.2	Login/Signup Page	74
	6.3	Dashboard	76
	6.4	Disease Prediction Page	77
	6.5	Medicine Recommendation Page	78
	6.6	Nutrition plan	79
	6.7	Workout Plan	80
	6.8	Disease Description Page	81
	6.10	Additional Interfaces	82-84

CHAPTER 1: INTRODUCTION

1. Background

Medicines are often referred to as the *lifeline of humanity*. They play a crucial role in saving lives, managing chronic illnesses, and improving the overall quality of human health. However, behind this life-saving role lies a less visible but equally important issue: the enormous wastage and unsafe disposal of medicines. Across the globe, including in India, millions of medicines remain unused each year, sitting in household cupboards, pharmacy shelves, and hospital storerooms until they eventually expire. This silent accumulation of unused medicines represents not only wasted resources but also lost opportunities to alleviate human suffering.

The problem becomes even more striking in a country like India, where healthcare accessibility is highly unequal. On one hand, urban households may discard unopened, unexpired medicines due to over-prescription, lack of adherence, or changes in treatment. On the other hand, countless families in rural and low-income communities struggle to afford even the most basic medications. This imbalance highlights a paradox: while medicines are wasted in one corner of society, another corner faces critical shortages. If channeled responsibly, these surplus medicines could make the difference between health and disease, or even life and death, for underprivileged populations.

Equally concerning is the environmental dimension of improper disposal. It is a common practice for households to throw expired tablets and syrups into domestic garbage bins or flush them into sinks and toilets. Once in the environment, the

chemical components of these drugs seep into soil and groundwater or enter rivers and lakes. Research has shown traces of antibiotics, hormones, and painkillers in water bodies, leading to harmful consequences such as contamination of the food chain and the rise of antimicrobial resistance (AMR). The World Health Organization (WHO) has already declared AMR a major global health threat, and irresponsible medicine disposal adds fuel to this growing crisis.

From an ecological perspective, the damage is long-lasting. Animals that consume contaminated water or waste can suffer toxic effects, while soil fertility is degraded by harmful chemical residues. Thus, pharmaceutical pollution not only threatens human health but also disrupts delicate environmental balances.

At the same time, the absence of structured systems for medicine redistribution in India means that enormous social opportunities are missed. Unlike some developed countries that have established medicine donation programs under strict regulatory frameworks, India lacks a centralized mechanism to collect, verify, and redistribute unexpired medicines at scale. As a result, hospitals, pharmacies, and households are left without safe, reliable options for what to do with their surplus stocks.

This project emerges as a response to these intertwined challenges of healthcare inequality, medicine wastage, and environmental harm. It proposes a **community-driven, technology-enabled platform** that bridges the gap between surplus and scarcity. By leveraging digital tools, the platform will allow individuals and institutions to donate unexpired medicines, guide users toward safe disposal centers, and create widespread awareness about responsible practices. More than just a technical solution, this initiative represents a movement toward a **culture of sharing, sustainability, and responsibility** in medicine usage.

The vision is not only to prevent waste but also to transform waste into value—unused medicines into relief for the needy, safe disposal into environmental preservation, and awareness into long-term behavioral change. In doing so, this project seeks to establish a model that addresses pressing social needs, protects the environment, and strengthens the very foundation of healthcare accessibility in India and beyond.

2. Objectives

The objectives of the project are as follows:

1. **Centralized Medicine Donation Platform:** Create a secure, user-friendly online platform where individuals and institutions can donate surplus, unexpired medicines, and where recipients can request the medicines they require.
2. **Geolocation-Based Disposal Mapping:** Integrate GPS-based mapping features to guide users toward the nearest safe medicine disposal centers, ensuring expired or unsafe medicines are not misused.
3. **Awareness and Education:** Build an educational hub within the platform that provides information about proper medicine storage, safe disposal methods, and the environmental risks of improper practices.
4. **Support for Vulnerable Groups:** Empower low-income communities, NGOs, and charitable healthcare institutions by providing them access to donated medicines, reducing their financial burden.
5. **Environmental Protection:** Promote eco-friendly practices by minimizing improper disposal of pharmaceuticals and ensuring compliance with safe waste management guidelines.

3. Purpose

The purpose of the project extends beyond technology—it aims to combine social impact, environmental responsibility, and healthcare accessibility:

- **Social Impact:** By channeling safe, unused medicines toward those in need, the project strengthens community health systems and provides relief to underprivileged groups who struggle to afford essential drugs.
- **Environmental Responsibility:** The project helps reduce pharmaceutical pollution by encouraging users to adopt proper disposal methods, preventing hazardous chemicals from entering ecosystems.
- **Public Engagement:** By educating the public about safe disposal and medicine-sharing practices, the platform builds awareness and encourages collective responsibility.
- **Healthcare Support:** The project provides pharmacies and hospitals with an alternative solution for managing surplus stock, reducing storage costs and preventing wastage.

4. Scope

The scope of this project defines its primary features, user roles, and initial implementation strategies:

- **User Roles:**
 - **Donors:** Individuals, pharmacies, or hospitals who donate surplus medicines.

- **Receivers:** Individuals in need of medicines or NGOs that distribute them.
- **Admins:** Moderators who manage platform activity, verify donations, and monitor compliance.
- **Core Features:**
 - Secure user registration and login system.
 - Dedicated pages for donors to list medicines and for receivers to request them.
 - GPS-based disposal site locator to direct users to nearby authorized collection centers.
 - Admin tools for moderation, approval, and generating reports.
 - Knowledge center with FAQs, guidelines, and awareness content.
- **Geographical Coverage:**

The project will initially be launched in selected Indian cities such as **Delhi, Mumbai, and Bengaluru**, where pilot studies can be conducted.
- **Scalability Plans:**

In future phases, the platform will expand to include:

 - Mobile applications for Android and iOS.
 - Multilingual support to reach diverse communities.
 - Partnerships with government health departments, NGOs, and pharmaceutical companies for scaling impact.

5. Applications

The proposed platform can have wide-reaching applications across various stakeholders:

1. **Individuals:** Users can easily donate unused medicines or request essential medicines without financial burden.
2. **Pharmacies and Hospitals:** Institutions can list surplus stock, manage inventory efficiently, and participate in socially responsible disposal practices.
3. **NGOs and Nonprofits:** Social organizations can access medicines for their community healthcare projects and donation drives.
4. **Government Agencies:** Authorities can use the platform to strengthen public health campaigns and monitor medicine waste trends.
5. **Environmental Organizations:** Agencies can track disposal data, identify pharmaceutical pollution hotspots, and contribute to environmental preservation.

6. Expected Achievements (Outcomes)

Upon successful implementation, the project is expected to deliver the following measurable outcomes:

- **Waste Reduction:** A tangible decrease in the amount of expired or wasted medicines at household and institutional levels.
- **Increased Accessibility:** Greater availability of essential medicines for low-income families and rural populations.
- **Public Awareness:** Enhanced community understanding of the importance of safe disposal and responsible medicine use.

- **Replicability:** The platform can serve as a **blueprint** for similar initiatives in other regions and countries.
- **Collaboration:** Stronger partnerships among individuals, NGOs, pharmacies, hospitals, and government agencies.
- **Scalability:** The foundation for developing long-term, self-sustaining networks for medicine donation and reuse.

CHAPTER 2: SURVEY OF TECHNOLOGIES

- **2.1 Frontend Technologies**
- 2.1.1 Comparison of Frontend Frameworks

◦ Aspect	◦ React	◦ Angular	◦ Vue.js
◦ Popularity	◦ Very High	◦ High	◦ Growing rapidly
◦ Learning Curve	◦ Moderate	◦ Steep	◦ Gentle
◦ Flexibility	◦ Highly flexible, component-based	◦ Opinionated, full framework	◦ Lightweight, flexible
◦ Tooling	◦ Rich ecosystem including Create React App	◦ Rich ecosystem including Create React App	◦ Smaller but mature tooling
◦ Community	◦ Large,	◦ Corporate-	◦ Growing, vibrant

	active	driven, large	
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- For your medicine donation platform, **React** is well-suited for building interactive user interfaces, such as dashboards, real-time updates, and reusable components like donation listings, request forms, and maps.
-
- 2.1.2 HTML5:
- HTML5 provides the semantic elements essential for structuring the content of web pages — forms for users to register, submit medicine donation details, and view listings.
- 2.1.3 CSS3
- CSS3 is used to style components, ensuring your platform is visually appealing and responsive. Techniques such as Flexbox, Grid Layout, and media queries enable a mobile-friendly design, critical for users accessing via smartphones.
- 2.1.4 JavaScript (ES6+)
- JavaScript enables front-end interactivity, including form validation, dynamic content updates, and communicating with back-end APIs for submitting donations, requests, and map data.
- 2.1.5 Bootstrap
- Bootstrap is a widely-used CSS framework offering pre-styled components such as buttons, forms, and navigation bars. It accelerates UI development and provides consistent styling, ensuring your site looks professional across devices.
- 2.1.6 Features of Frontend Stact:
 - **Responsiveness** for cross-device compatibility.
 - **Component Reusability** to build modular UI.
 - **State Management** via libraries like Redux or Context API for efficient data handling.
 - **Integration with APIs** for real-time updates and interactive features.
- **Map Integration** (Google Maps or OpenStreetMap API) enables location-

based browsing of donations and disposal sites.

○ 2.1.7 Advantages:

- Rapid development with React and Bootstrap.
- Easy maintainability through component-based architecture.
- Responsive design improves accessibility.
- Large community & resources for troubleshooting and enhancements.

○ 2.1.8 Disadvantages

- Initial learning curve for React and state management libraries.
- Heavy reliance on JavaScript; users with JS disabled will have degraded experience.
- Potential overhead with large-scale state synchronization if not managed properly.

-

○ 2.1.9 Summary of Frontend Technologies

○ TECHNOLOGY	○ PURPOSE	○ ADVANTAGE	○ DISADVANTAGE
○ HTML5	○ Structure content	○ Universal support SEO-friendly	○ Static without JS
○ CSS3	○ Styling	○ Responsive, customization	○ Complexity with scale
○ JavaScript	○ Interactively	○ Dynamic user experience	○ Can become complex
○ React	○ UI components	○ Reusable efficient rendering	○ Requires build tooling
○ Bootstrap	○ Styling	○ Fast	○ Overused UI

	framework	prototyping, consistency	look
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2.2 Backend Technologies

- 2.2.1 Node.js with Express.js
 - *Node.js* enables server-side JavaScript execution, suitable for asynchronous handling of API requests.
 - *Express.js* is a minimalist framework simplifying routing, middleware handling, and API development.
 - This stack efficiently supports real-time features like live updates of donation statuses.
- 2.2.2 Database: MongoDB
 - A NoSQL document database well-suited for flexible data models like donations, user profiles, and location data.
 - Enables rapid development with JSON-like storage, horizontal scalability, and geospatial queries (for mapping disposal points)
- 2.2.3 Authentication and Authorization
 - Use of *JWT (JSON Web Tokens)* for stateless user session management.
 - Implement role-based access: donors, receivers, admins.
- 2.2.4 Geolocation and Mapping APIs
 - Google Maps API or alternatives for displaying donation sites and disposal locations.
 - Reverse geocoding and geospatial queries enable searching donations nearby.
- 2.2.5 Additional Technologies:
 - *RESTful APIs* or *GraphQL* facilitate efficient data exchange between frontend and backend.
 - *Cloud Hosting:* AWS, Heroku or DigitalOcean for deployment.
 - *Notification Service:* Email or push notifications to alert donors or receivers.
- 2.2.6 Summary of Backend Technologies

◦ Techonology	◦ purpose	◦ Advantage s	◦ Disadvantage s
◦ Node.js+Express. js	◦ API & server	◦ High performan ce JS eco- system	◦ Single threaded(limits cpu- heavy tasks)
◦ MongoDB	◦ DATABASE	◦ Flexible schema , geospatial queries	◦ Lacks relational integrity as in SQL
◦ JWT	◦ Authenticatio n	◦ Stateless scalable	◦ Token management complexity
◦ Google maps API	◦ Geolocation & mapping	◦ Rich features , high accuracy	◦ Cost for high usage

Chapter 3 – REQUIREMENT ANALYSIS

3.1 Problem Definition

In many households, unused and unexpired medicines often go to waste, while at the same time, there are people who cannot afford or access the medicines they need. This leads to two major problems:

Wastage of medicines due to lack of proper redistribution mechanisms.

Lack of accessibility for patients in need, especially in rural or low-income areas. Existing healthcare platforms focus mainly on **prescriptions, pharmacy sales, or recommendations**, but very few address the **medicine exchange ecosystem**.

There is no structured system that ensures safe, verified, and efficient exchange of medicines between users.

The proposed **Medicine Exchange Platform** aims to bridge this gap by creating a secure, transparent, and user-friendly online system where:

Users can **list medicines** they want to donate/exchange.

Interested users can **search and request medicines**.

The system ensures **safety, verification, and proper communication** between parties.

3.2 Requirement Specification

A) Functional Requirements:

User Registration & Authentication: Secure login and signup using email/mobile.

Medicine Listing Module: Users can upload details of medicines (name, expiry, quantity, manufacturer, condition).

Search & Filter: Users can search medicines by name, category, or availability.

Medicine Request & Exchange:

Option to request medicine from another user.

System manages communication between donor and receiver.

Verification System: Medicines must meet criteria (not expired, proper packaging).

Notifications: Users receive alerts for medicine requests, approvals, or updates.

History & Tracking: Dashboard to track listed medicines, requests, and exchanges.

Reporting/Flagging: Users can report unsafe or fraudulent activities.

B) Non-Functional Requirements:

Security: Encrypted storage of user data, safe communication channels.

Scalability: Platform should handle a large number of users and medicines.

Performance: Response time < 3 seconds.

Usability: Simple, intuitive interface suitable for all age groups.

Compatibility: Accessible via web and mobile (Android/iOS).

Reliability: Ensure minimal downtime and high availability.

Data Validation: Prevent expired or duplicate medicines from being listed.

3.3.2 Scheduling Process:

Task Sequencing:

Requirements gathering → UI/UX design → Backend & database setup → Medicine listing & search → Exchange request & approval system → Notifications → Testing → Deployment.

Task Estimation:

Requirement Analysis: 2 weeks

UI/UX Design: 3 weeks

Backend & Database Development: 4 weeks

Medicine Exchange Module: 4 weeks

Notifications & Tracking: 2 weeks

Testing: 3 weeks

Deployment: 1 week

Task Dependencies:

Database design must be completed before backend development.

Exchange module depends on listing & search module.

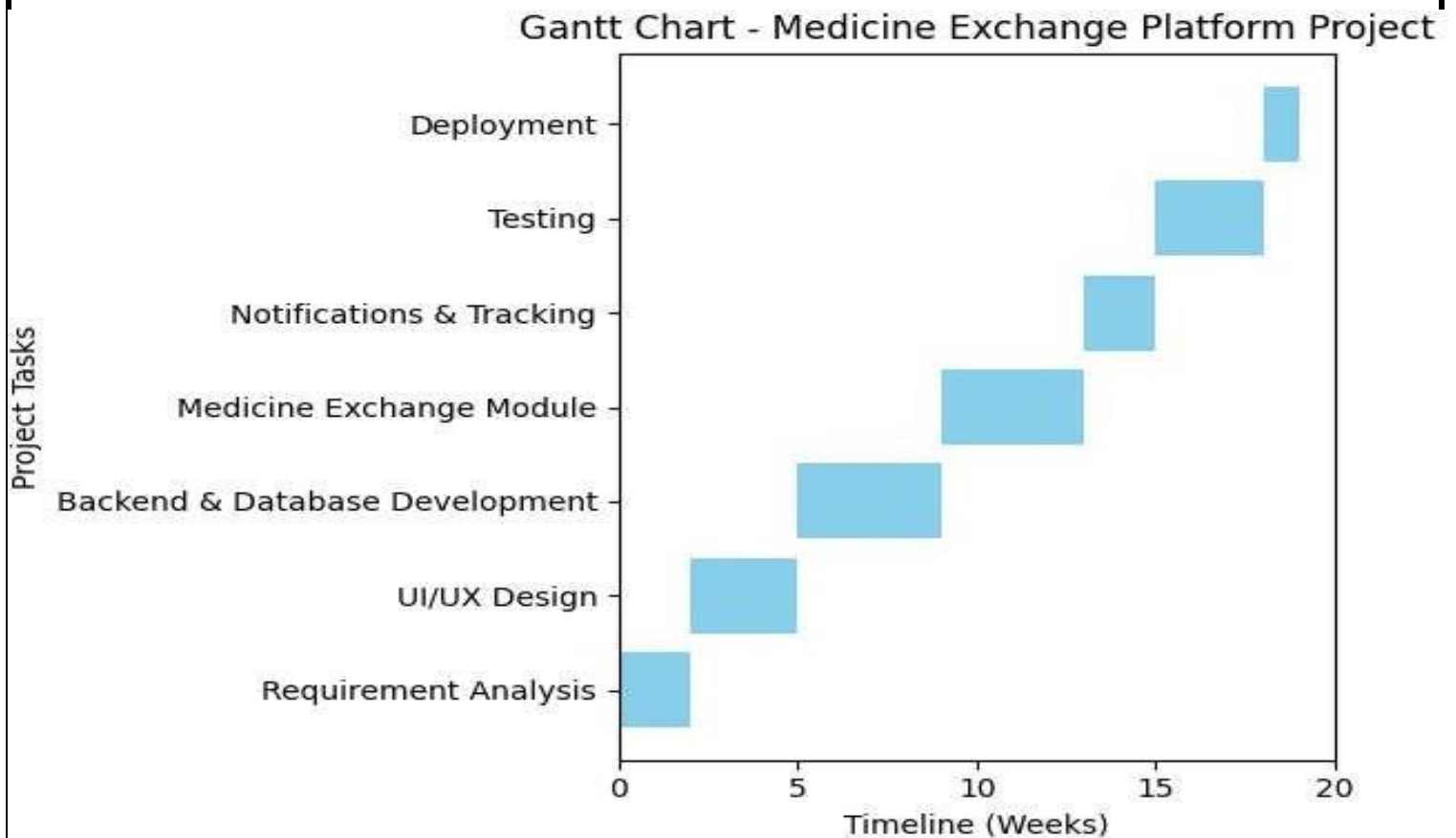
Notifications depend on exchange module.

Review and Adjustment:

Agile methodology ensures incremental progress.

Weekly sprint reviews allow adjustments for delays or changes in scope.

3.3.3 Techniques Used for Scheduling – Gantt Chart:



The Gantt chart will be used to track timelines, dependencies, and progress of modules visually.

3.4.1 Developer Side:

Development Environment: VS Code / PyCharm.

Programming Languages & Frameworks:

Frontend: HTML, CSS, JavaScript, Bootstrap/React.

Backend: Node.js.

Database Management System:

MySQL / PostgreSQL for structured storage of medicines and users.

Testing Tools: Postman (API testing), Selenium (UI testing).

Operating System: Cross-platform (Windows/Linux/Mac).

3.4.2 Client Side:

Web browsers: Chrome, Firefox, Edge, Safari.

Mobile access: Android/iOS browser or mobile app.

Internet: Minimum 1 Mbps connection.

3.5 Hardware Requirement:

3.5.1 Developer Side:

Processor: Intel i5 (min) / i7 (recommended).

RAM: 8 GB (min), 16 GB recommended.

Storage: 256 GB SSD (min).

GPU: Not mandatory, unless AI/ML-based verification is added.

Internet: High-speed for testing and deployment.

3.5.2 Client Side:

PC/Laptop: Dual-core processor, 4 GB RAM, standard browser.

Mobile Devices: Android 8.0+ or iOS 11+.

Internet: 3G/4G minimum.

3.6 Preliminary Product Description:

The **Medicine Exchange Platform** provides a secure online system for donating, sharing, or exchanging medicines.

It enables users to:

List available medicines with details.

Search and request medicines from other users.

Track medicine donations/exchanges via a personalized dashboard.

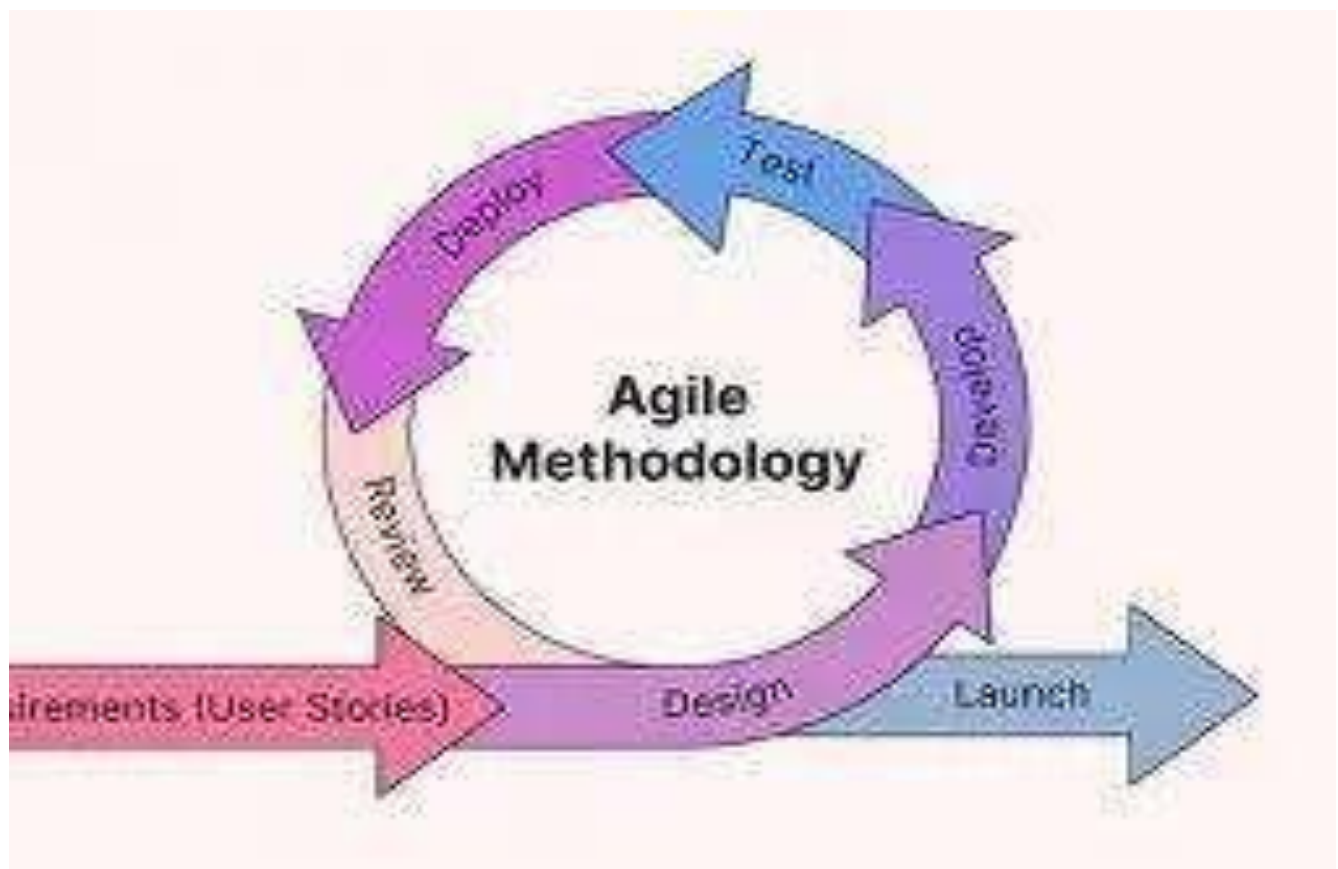
Ensure safety with an in-built verification and reporting system.

The platform not only reduces **medicine wastage** but also improves **accessibility for patients in need**, contributing to sustainable healthcare.

3.7 Conceptual Models:

SDLC Model: Phases include Requirement Analysis, System Design, Development, Testing, Deployment, and Maintenance.

Agile Model: Adopted for incremental delivery. Each sprint (2–3 weeks) focuses on one core feature (e.g., listing, exchange, notifications). This ensures continuous feedback, flexibility, and better user satisfaction.



CHAPTER 4:SYSTEM DESIGN

4.1 Module Description

4.1.1 User Authentication Module

- **Purpose:** Securely manage user registration, login, and role-based access (e.g., donor, buyer, admin).
 - **Features:**
 - Email/Password or OAuth login.
 - Password encryption (e.g., bcrypt).
 - Session management using JWT.
 - Role-based dashboards (donor dashboard vs buyer dashboard).
-

4.1.2 Symptom Input and Analysis Module

- **Purpose:** Allow users to input symptoms and get possible condition insights.
 - **Features:**
 - Interactive form for symptom entry.
 - Rule-based or AI-driven symptom analysis.
 - Suggest if consultation with a doctor is necessary.
 - Provides **non-diagnostic guidance only** (to stay safe legally).
-

4.1.3 Personalized Recommendation Module

- **Purpose:** Recommend medicines, health practices, or donation opportunities based on user history and needs.
 - **Features:**
 - Suggest medicines frequently bought/donated.
 - Personalized donation prompts (e.g., "You have surplus unexpired medicine, would you like to donate?").
 - Health suggestions linked with symptoms entered.
-

4.1.4 Optical Character Recognition (OCR) Module

- **Purpose:** Extract medicine details from prescriptions or medicine strips using OCR.
 - **Features:**
 - Upload prescription image.
 - Extract drug name, dosage, and expiry date.
 - Validate against database of medicines.
 - Auto-fill medicine donation form.
-

4.1.5 Medication Information and Interaction Module

- **Purpose:** Provide safe usage information and highlight drug interactions.
- **Features:**
 - Display dosage, side effects, manufacturer details.
 - Warn about possible drug interactions (e.g., Drug A + Drug B not safe).
 - Suggest alternatives for unavailable or unsafe medicines.
 - Link to reliable medical sources (e.g., WHO, FDA).

4.1.6 User Dashboard Module

- **Purpose:** Provide a central space where users can view and manage their activities (purchases, donations, symptoms, recommendations, etc.).
- **Features:**
 - **Buyers:** View purchased medicines, track orders, access recommendations.
 - **Donors:** Manage donated medicines, check donation history.
 - **General Users:** View symptom history, personalized recommendations.
 - **Admins:** Manage users, medicines, and platform analytics.
- **Key Components:**
 - Order/Donation history section.
 - Notifications and reminders (e.g., “Medicine expiry approaching”).
 - Quick access to modules (Symptom Input, OCR upload, etc.).

4.2 Database Design

Field Name	Data Type	Description
user_id (PK)	INT (AUTO)	Unique user ID
name	VARCHAR(100)	Full name of the user

email	VARCHAR(150)	User's email (unique)
password_hash	VARCHAR(255)	Encrypted password
role	ENUM('buyer','donor','admin')	User role
created_at	DATETIME	Account creation timestamp

4.2.2 Symptoms Table Design

Field Name	Data Type	Description
recommendation_id (PK)	INT (AUTO)	Unique ID for recommendation
user_id (FK)	INT	Linked to Users table
recommendation_text	VARCHAR(255)	Recommendation message
type	ENUM('medicine','donation','health')	Category
created_at	DATETIME	Timestamp

4.2.4 OCR Data Table Design

Field Name	Data Type	Description
ocr_id (PK)	INT (AUTO)	Unique OCR record ID
user_id (FK)	INT	Linked to Users table
image_path	VARCHAR(255)	Path/URL of uploaded image
extracted_text	TEXT	Raw extracted text
medicine_name	VARCHAR(100)	Identified medicine
expiry_date	DATE	Extracted expiry date
expiry_date	DATETIME	Timestamp

4.2.5 Medication Information & Interaction Table Design

Field Name	Data Type	Decrption
med_id (PK)	INT (AUTO)	Unique medicine ID
medicine_name	VARCHAR(100)	Medicine name
details	TEXT	Usage details, side effects
interaction_with	VARCHAR(100)	Other drug name (if interaction exists)
interaction_effect	VARCHAR(255)	Effect of interaction
source_reference	VARCHAR(255)	Link/reference for info

4.2.6 Data Integrity and Constraints

To ensure correctness, consistency, and reliability of stored data:

- **Primary Keys & Foreign Keys:** Every table has a unique primary key; foreign keys maintain relationships (e.g., `user_id` in Symptoms, OCR, Recommendations).
- **Constraints:**
 - NOT NULL for required fields (e.g., `email`, `medicine_name`).
 - UNIQUE constraint for `email` in Users table.
 - CHECK constraints for expiry date (must be greater than current date).
 - ENUM constraints for roles (`buyer`, `donor`, `admin`) and recommendation types.
- **Referential Integrity:** Cascade updates/deletes where appropriate.
- **Validation Rules:** OCR-extracted medicines cross-checked against medication database before storage.

4.3 Procedural Design

4.3.1 Logic Diagrams

4.3.1.1 Use Case Diagram

- **Actors:** User (buyer/donor), Admin, System.

- **Use Cases:**

- Register/Login.
- Purchase medicine.
- Donate medicine.
- Input symptoms → Get analysis.
- Upload prescription (OCR).
- Get personalized recommendations.
- View dashboard.
- Admin manages users & medicines.

4.3.1.2 Activity Diagram (Example: Donation Flow)

1. User logs in.
2. Navigates to “Donate Medicine.”
3. Enters details / uploads medicine strip via OCR.
4. System validates expiry.
5. Donation record stored in database.
6. User gets confirmation → Dashboard updated.

4.3.1.3 Sequence Diagram (Example: Purchase Flow)

1. **User** → selects medicine.
2. **System** → checks stock in DB.
3. If available → adds to cart.
4. User confirms purchase.
5. System deducts quantity → updates DB.
6. Payment gateway interaction (if implemented).
7. Confirmation sent → Dashboard updated.

4.3.1.4 Entity Relationship Diagram (ERD)

Entities:

- **Users** (user_id, name, email, role, ...)
- **Symptoms** (symptom_id, user_id, symptom_text, ...)
- **Recommendations** (recommendation_id, user_id, recommendation_text, ...)
- **OCR Data** (ocr_id, user_id, medicine_name, expiry_date, ...)

- **Medicines/Interactions** (med_id, medicine_name, interaction_with, ...)

Relationships:

- Users **1**—**M** Symptoms.
- Users **1**—**M** Recommendations.
- Users **1**—**M** OCR Data.
- Medicines **M**—**M** Interactions.

4.4 User Interface Design

- **Login/Signup Page** (secure access).
- **Dashboard** (different views for donor, buyer, admin).
- **Medicine Purchase Page** (search, filter, cart).
- **Donation Form with OCR Upload** (auto-fills fields).
- **Symptom Input Page** (interactive form).
- **Recommendation Page** (personalized suggestions).
- **Admin Panel** (manage users, medicines, analytics).

4.5 Security Issues

- **Authentication & Authorization:** JWT/session tokens.
- **Data Protection:** Password hashing (bcrypt), HTTPS/TLS for all comms.
- **Input Validation:** Prevent SQL injection, XSS.
- **File Uploads:** Sanitize OCR uploads (only allow images/PDF).
- **Access Control:** Role-based (admin vs donor vs buyer).
- **Audit Logs:** Track suspicious activities.

4.6 Test Cases Design:

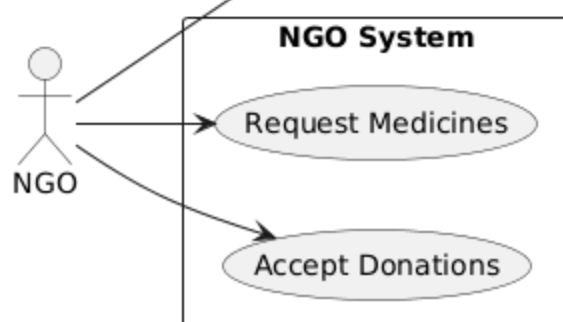
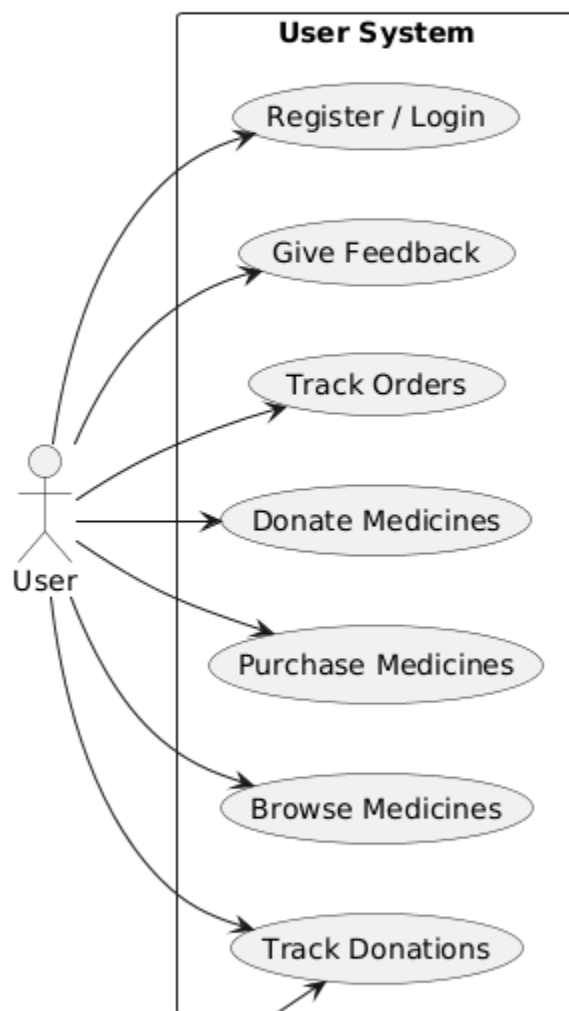
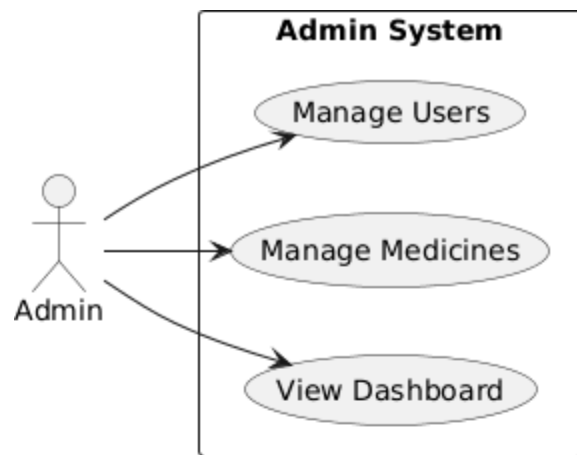
Test Cas	Description	Input	Expected Output	Actual Output	Status
e ID					
TC01	User login with valid credentials	Email + Password	Redirect to dashboard	Dashboard loads	pass
TC02	Login with invalid password	Email + Wrong Password	Show error	Error shown	pass
TC03	Add new donation	Medicine details	“Donation successful” message	Success message	pass
TC04	Purchase out-of-stock medicine	Select medicine with 0 qty	“Out of stock” Warning	Warning shown	pass
TC05	Upload prescription (OCR)	Image file	Extracted text shown	Text extracted	pass
TC06	Symptom input	“Fever, headache”	Recommendation generated	Recommendation displayed	pass

4.3 Procedural Design

4.3.1 Logic Diagrams

4.3.1.1 Use case diagram.

- Use case Diagram for developer side and user.



.3.1.2 Activity diagram

- Activity Diagram:

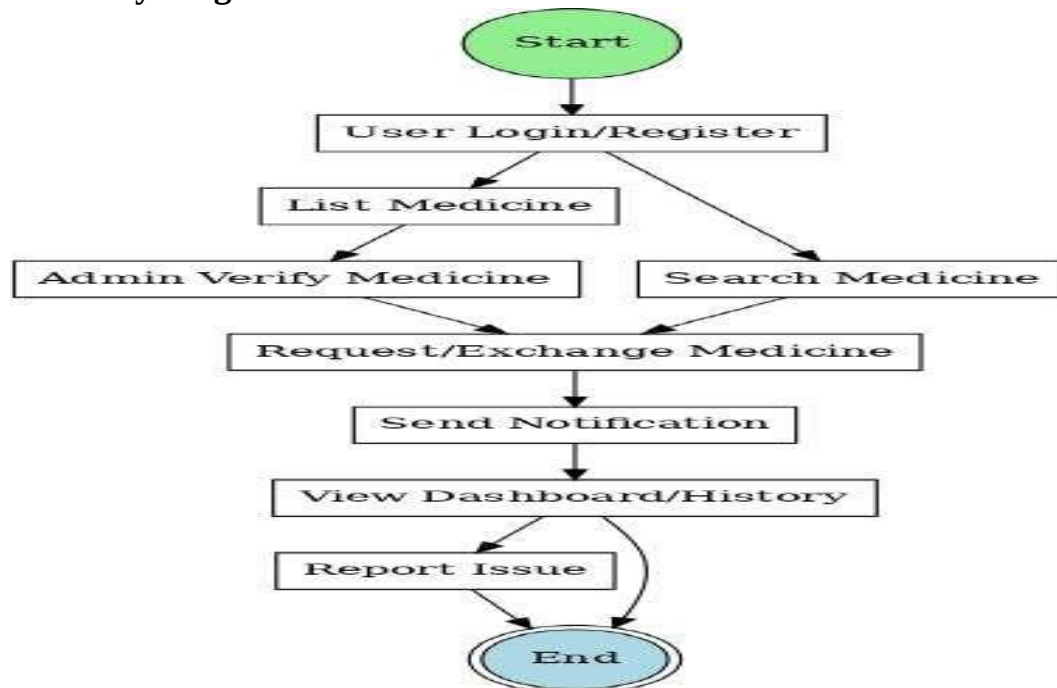


Fig. 4.2: Activity Diagram

4.3.1.3 Sequence diagram

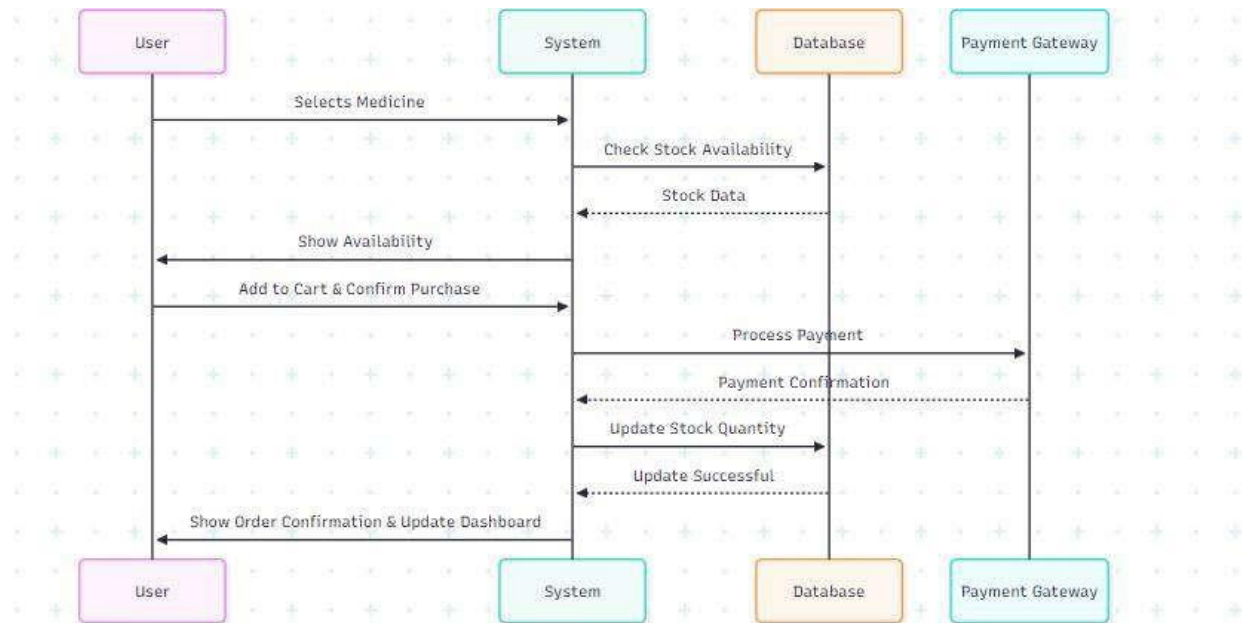


Fig. 4.3: Sequence Diagram

4.3.1.3 Entity Relationship(ER) diagram

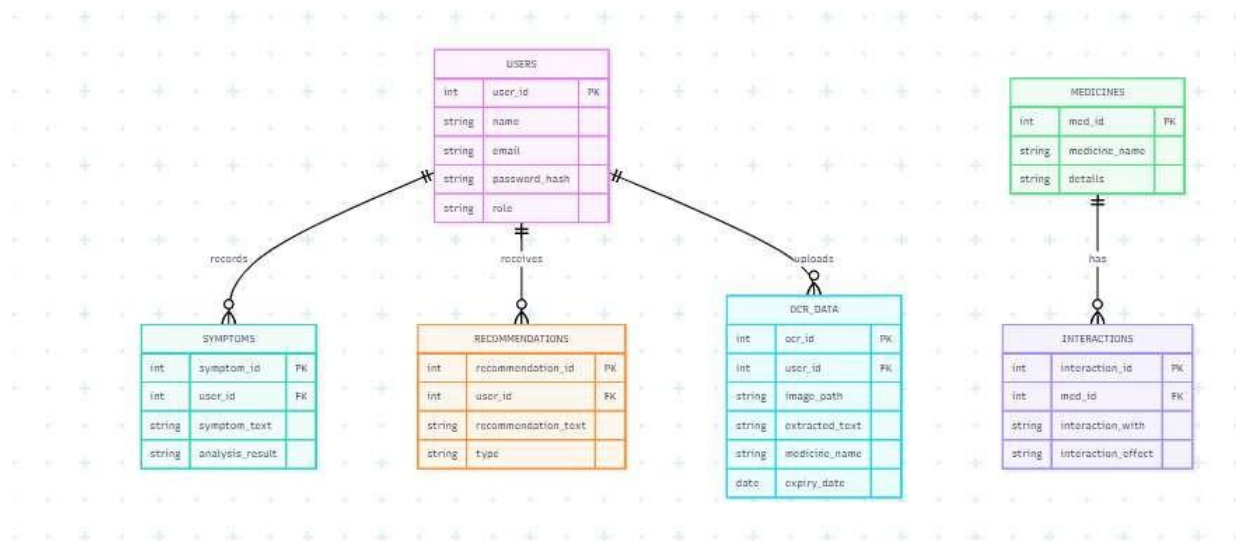


Fig. 4.4: Entity Relationship Diagram

Login page:



[Home](#)
[Browse](#)
[Sell](#)
[Scan](#)
[Alternates](#)
[Disposal](#)
[Profile](#)
[Login](#)

Log in

Enter your credentials to access your account.

Email

Password

[Log in](#)
[Create account](#)

Regrestration page:

Create an account

Join MediConnect to buy, sell, and dispose of medicines safely.

Name

Email

I am a...

Password

[Sign up](#) [Have an account? Log in](#)

Dashboard :

Home Browse Sell Scan Alternates Disposal Profile [Logout \(Admin\)](#)

paracetamol
urvi • Exp: 32154-03-20 • Qty: 100
VACCINE ₹100000000
[Buy](#)

Paracure 650 mg Tablet
MediLife Pharmaceuticals • Exp: 2029-11-15 • Qty: 1
GENERAL ₹2.5
[Buy](#)

AmoxiCare 500mg
HealthPharm • Exp: 2026-05-01 • Qty: 1
ANTIBIOTIC ₹9.99
[Buy](#)

Chapter 5: Implementation and Testing

5.1 Implementation Approaches

The Unused Medicine Exchange and Medical Waste Management System was implemented using a full-stack web development approach. The backend was developed using Node.js with Express.js, while data persistence was handled through a lightweight JSON-based database. The system follows a modular architecture to ensure maintainability, scalability, and clarity.

5.1.1 Backend Initialization (Express.js Setup)

```
const express = require('express');
const bodyParser = require('body-parser');
const cors = require('cors');
const app = express();
const PORT = 3000;

// Middleware
app.use(cors());
app.use(bodyParser.json({ limit: '50mb' }));
app.use(express.static('public'));
```

Description:

The backend server is initialized using Express.js, a lightweight and efficient web application framework for Node.js. Express was chosen due to its simplicity and flexibility in handling RESTful APIs.

The following middleware components are configured:

- **CORS (Cross-Origin Resource Sharing):** Enables secure communication between frontend and backend running on different ports or domains.
- **Body Parser:** Configured with a 50mb payload limit to support image uploads (e.g., medicine packaging or prescription photos).

- **Static File Serving:** Allows frontend assets to be served directly from the `public` directory.

The server runs on port 3000 and handles all API routes related to authentication, medicines, disposal, and payment modules.

5.1.2 Database Configuration (JSON-Based Persistence)

```
const DB_FILE = './db.json';

const readDB = () => {
  try {
    return JSON.parse(fs.readFileSync(DB_FILE));
  } catch (e) {
    return { users: [], medicines: [], orders: [],
disposals: [] };
  }
};

const writeDB = (data) =>
  fs.writeFileSync(DB_FILE, JSON.stringify(data, null,
2));
```

Description:

The system uses a JSON-based local database (`db.json`) for data persistence. This lightweight approach was selected for rapid prototyping and simplicity.

The database contains the following collections:

- `users`
- `medicines`
- `orders`
- `disposals`

Two utility functions manage database operations:

- `readDB()` – Reads and parses JSON data from the file.
- `writeDB()` – Writes updated data back into the file with structured formatting.

This design ensures:

- Data consistency across server restarts
 - Simplicity in debugging
 - Easy scalability for future migration to MongoDB or SQL
-

5.1.3 User Authentication with OTP Logic

```
const otpStore = {}; // Memory Store for OTPs

// LOGIN ENDPOINT
app.post('/api/login', (req, res) => {
  const { email, password } = req.body;
  const db = readDB();
  const user = db.users.find(u =>
    u.email === email && u.password === password
  );

  if (!user)
    return res.status(401).json({ error: 'Invalid
credentials' });

  const otp = Math.floor(100000 + Math.random() *
900000).toString();

  otpStore[email] = {
    otp,
    user,
    expires: Date.now() + 300000
  };

  res.json({
    success: true,
```

```

        otpRequired: true,
        message: `OTP generated: ${otp}`
    });
});

```

Description:

The authentication mechanism follows a two-step verification model:

1. **Credential Validation** – Email and password are verified against stored user records.
2. **OTP Generation** – A 6-digit One-Time Password is generated and stored temporarily in memory with a 5-minute expiry.

Key features:

- OTP expiration logic (300,000 ms = 5 minutes)
- Secure login simulation similar to banking systems
- Prevention of unauthorized access
- Rejection of expired or invalid OTPs during verification

This approach enhances system security and mimics real-world authentication flows.

5.1.4 Medicine Marketplace Implementation (Sell Module)

```

app.post('/api/medicines', (req, res) => {
    const med = {
        id: Date.now(),
        ...req.body,
        created: new Date().toISOString()
    };

    const db = readDB();
    db.medicines.unshift(med);
    writeDB(db);

    res.json({ success: true, medicine: med });
});

```

Description:

This endpoint enables users to upload unused medicines into the marketplace.

Each medicine entry contains:

- Unique ID (timestamp-based)
- Name
- Category
- Price
- Expiry date
- Uploaded image (if provided)
- Seller information
- Creation timestamp

The `unshift()` method ensures that newly uploaded medicines appear at the top of the marketplace listing.

The module supports:

- Circular economy participation
- Reduction of medical waste
- Affordable medicine redistribution

5.1.5 Medical Waste Disposal Tracking

```
app.post('/api/disposal', (req, res) => {
  const reqData = {
    id: Date.now(),
    status: 'Pending',
    created: new Date().toISOString(),
    ...req.body
  };

  const db = readDB();
  db.disposals.unshift(reqData);
  writeDB(db);
});
```

```
    res.json({ success: true, request: reqData });
  });
```

Description:

The disposal module allows users to submit requests for safe medical waste collection.

Features include:

- Automatic "Pending" status assignment
- Timestamp tracking
- Persistent storage in database
- Status upgradation workflow (Pending → In Progress → Completed)

This ensures proper tracking and safe disposal of expired or unsafe medicines.

5.1.6 Payment Processing and Order Confirmation

```
app.post('/api/pay', (req, res) => {
  const { userId, amount, method } = req.body;

  setTimeout(() => {
    const order = {
      id: Date.now(),
      userId,
      amount,
      status: 'Paid',
      date: new Date().toISOString()
    };

    const db = readDB();
    db.orders.unshift(order);
    writeDB(db);

    console.log(
      `[SMS SYSTEM] Sending: "Order #${order.id}"`
    );
  }, 1000);
});
```

```

confirmed. Paid: ₹${amount}"`
    );

    res.json({
      success: true,
      orderId: order.id,
      message: 'Payment Successful'
    });

  }, 1500);
});

```

Description:

The payment module simulates integration with an external payment gateway.

Key functionalities:

- Supports multiple payment methods (UPI/Card simulation)
- Simulated processing delay (1.5 seconds)
- Order persistence after successful payment
- Mock SMS confirmation via console log

This module completes the end-to-end buying cycle in the marketplace.

5.2 Testing Approach

The system was tested using a combination of Integration Testing (IT) and Unit Testing (UT). All APIs were validated using tools such as Postman to ensure correct request-response behavior and database synchronization.

Testing focused on:

- Authentication security
- Data persistence
- Marketplace workflow
- Disposal tracking

- Payment lifecycle
- OTP expiration validation

Test ID	Module	Objective	Expected Outcome	Status
IT-AUTH-001	POST /api/signup	Verify user registration and uniqueness	User created in db.json; duplicate emails rejected	Pass
IT-AUTH-002	POST /api/login	Validate password and OTP generation	Returns login success and 6-digit OTP	Pass
UT-OTP-003	verify-login-otp	Ensure expired/wrong OTPs are rejected	System returns error for incorrect/expired OTP	Pass
IT-MED-004	POST /api/medicines	Test medicine listing in marketplace	Medicine appears in global and user history	Pass
IT-PAY-005	POST /api/pay	Validate payment lifecycle	Order recorded as 'Paid'; SMS log triggered	Pass
IT-DISP-006	POST /api/disposal	Verify disposal request persistence	Data saved with 'Pending' status	Pass
UT-DB-007	Database Sync	Ensure persistence on restart	Data remains intact after server reboot	Pass

•

5.2.1 Test Cases

5.2.2 Testing Results Summary

All modules were tested successfully under normal operating conditions. Edge cases such as:

- Duplicate registration attempts
- Incorrect login credentials
- Expired OTP verification
- Server restart data retention
- Payment simulation delays

were handled correctly.

The system demonstrated:

- Stable API response behavior
- Consistent database synchronization
- Reliable status tracking
- Secure authentication logic

5.3 Advanced Module Implementation

5.3.1 Admin Dashboard and Approval Workflow

The system includes a role-based access control mechanism where users are categorized as either:

- **Normal User**
- **Administrator**

The Administrator has elevated privileges, including:

- Viewing all registered users
- Managing medicine listings
- Approving or rejecting uploaded medicines
- Updating disposal request statuses
- Monitoring order transactions

Medicine Approval Logic

When a user uploads a medicine, the system assigns it an initial status:

status: "Pending Approval"

The Admin reviews the listing and updates the status to:

- "Approved" – Visible in marketplace
- "Rejected" – Removed from public listing

This ensures:

- Prevention of counterfeit medicines
- Expiry date verification
- Quality control compliance

This moderation layer increases trust and platform credibility.

5.3.2 Medicine Status Tracking System

Each medicine in the system maintains a lifecycle status:

- Pending Approval
- Available
- Requested
- Sold
- Expired
- Rejected

Status is updated automatically or manually based on system events.

Example:

- When payment succeeds → status changes to "Sold"
- If expiry date < current date → status automatically becomes "Expired"

This structured workflow ensures real-time tracking and transparency.

5.3.3 Expiry Date Validation Logic

The system includes automated expiry validation to prevent unsafe distribution.

Logic Concept:

1. On medicine upload:
 - Expiry date must be greater than current date.
2. If expiry date equals current date:
 - Marked as "Expiring Soon"
3. If expiry date is in the past:
 - Automatically rejected

Pseudo-validation logic:

```
const today = new Date();
const expiry = new Date(req.body.expiryDate);

if (expiry < today) {
    return res.status(400).json({ error: "Medicine expired"
});
}
```

This ensures compliance with pharmaceutical safety standards.

5.3.4 Search and Filtering Algorithm

The marketplace supports dynamic searching and filtering.

Users can filter medicines by:

- Medicine name
- Category
- Price range
- Expiry date
- Location (City-based)

Search is implemented using array filtering from the database:

- ```
const filtered = db.medicines.filter(m =>
 m.name.toLowerCase().includes(searchTerm.toLowerCase())
);
```

The algorithm ensures:

- Case-insensitive search
- Fast filtering for small-to-medium datasets
- Real-time results

This improves user experience and marketplace usability.

---

### 5.3.5 Location-Based Filtering (Latitude & Longitude)

To support local medicine exchange, the system integrates location filtering.

Process:

1. User enters city name.
2. City is converted into latitude and longitude (via geocoding API).
3. Medicine listings are filtered based on matching city or nearby radius.

Conceptual Logic:

```
if (medicine.city === user.city) {
 return true;
}
```

Future scalability allows radius-based calculations using distance formulas (Haversine Algorithm).

Benefits:

- Encourages local exchange
- Reduces transportation cost
- Supports sustainable circular economy model

---

### 5.3.6 Image Upload Handling

The system allows users to upload medicine images for verification.

Features:

- Base64 image handling
- Increased body size limit (50mb)
- Stored directly in JSON database

This enables:

- Visual verification
  - Reduced fraud risk
  - Better buyer confidence
- 

## 5.4 Security and Performance Considerations

### 5.4.1 Security Measures

The system incorporates several security features:

- OTP-based two-factor authentication
- Expiry validation checks
- Admin moderation workflow
- Controlled payment simulation
- Input validation at API level

Future enhancements may include:

- Password hashing (bcrypt)
  - JWT authentication
  - Rate limiting
  - HTTPS enforcement
-

### 5.4.2 Performance Considerations

Although JSON-based storage is used for prototyping, performance remains stable for:

- Small to medium user bases
- Limited concurrent access
- Academic demonstration purposes

For production-level deployment, migration to:

- MongoDB
- PostgreSQL
- Cloud storage

would be recommended.

---

## 5.5 Overall Implementation Summary

The Unused Medicine Exchange System successfully integrates:

- Secure authentication with OTP
- Marketplace module
- Payment simulation
- Disposal management
- Admin moderation
- Search and filtering
- Expiry validation
- Status tracking
- Data persistence

The implementation aligns with circular economy principles by:

- Reducing pharmaceutical waste
- Promoting responsible redistribution
- Enabling safe disposal tracking
- Encouraging sustainable healthcare practices

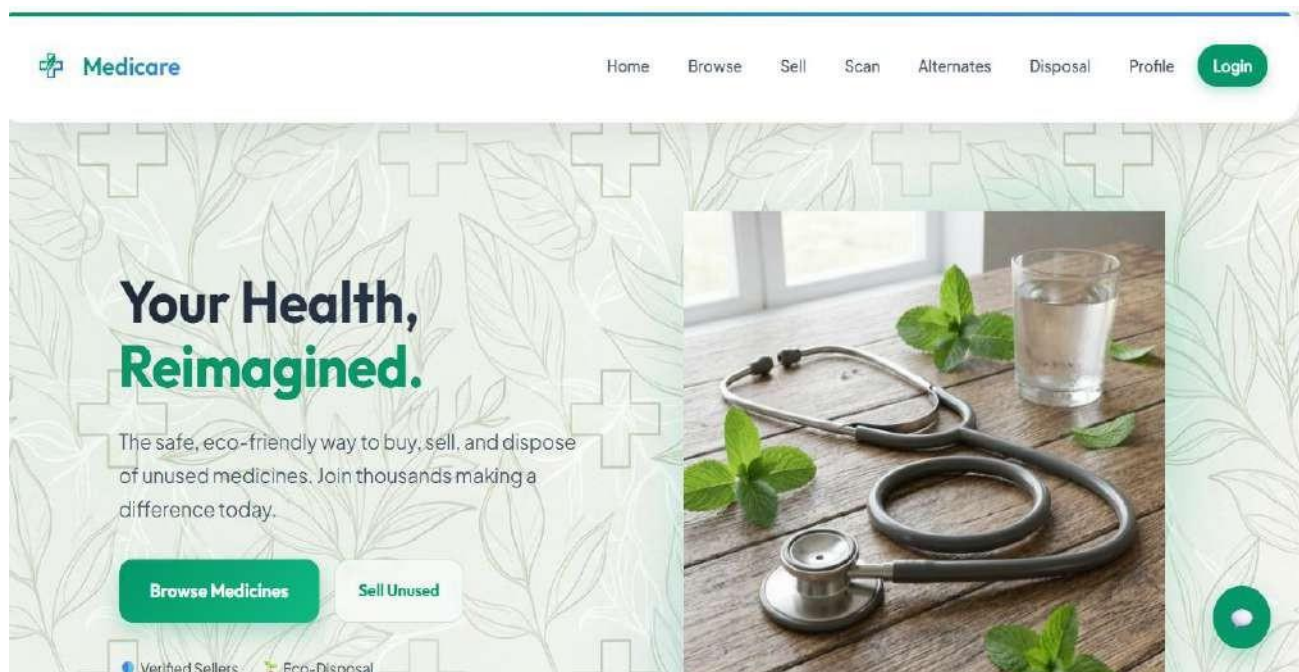
# CHAPTER 6 :IMPLEMENTATION AND INTERFACE DESIGN

## 6.1 Overview

This chapter explains the implementation details and interface design of the **Unused Medicine Exchange System**. The system has been developed by following modern web development practices with a focus on usability, performance, security, and sustainability. The chapter covers the overall design philosophy, technical architecture, user interfaces, and intelligent modules integrated into the system.

---

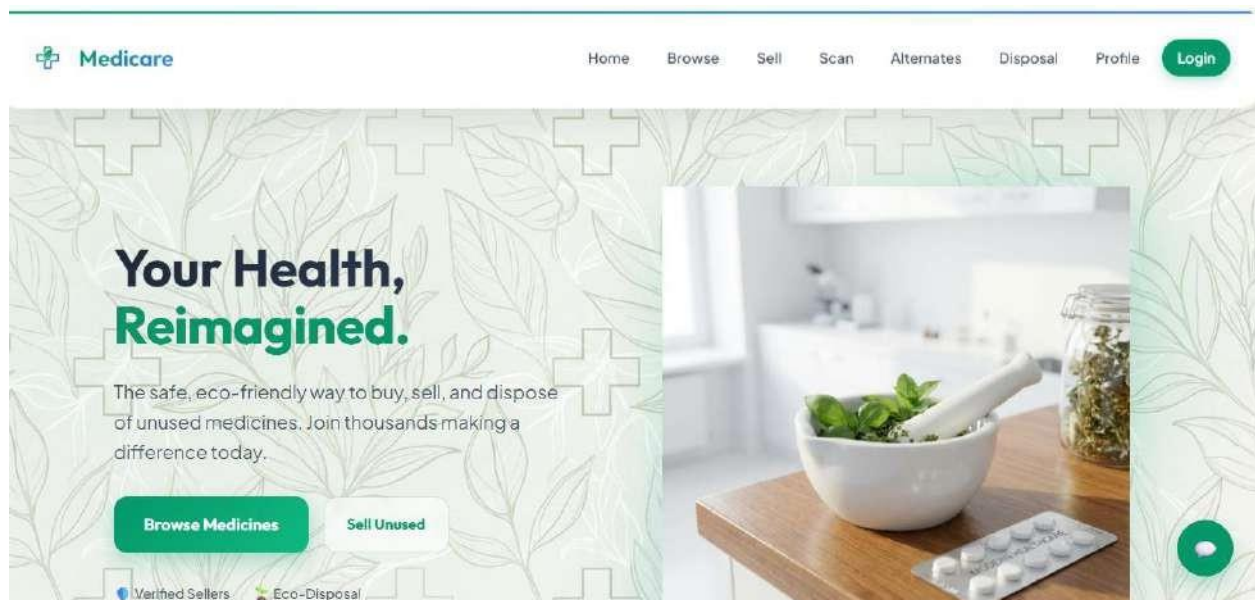
## 6.2 Implementation and Interface Design (Overview)



The implementation of the Unused Medicine Exchange System follows a **Core-First design philosophy**, where system functionality and performance are prioritized before visual enhancements. This approach ensures that essential features such as medicine exchange, disposal

scheduling, authentication, and recommendation logic operate reliably across all platforms.

## 6.2.1 Aesthetic Direction



The user interface adopts a **Vibrant Eco design palette** to align with the sustainability goals of the project. Emerald Green is used to represent environmental responsibility, while Blue is used to convey medical trust and reliability. This balanced color scheme helps build user confidence and enhances the overall visual appeal of the platform.

## 6.2.2 Design System

A **Glassmorphism-inspired user interface** is implemented to provide a modern and premium experience. Translucent cards, soft shadows, and subtle blur effects are applied to headers and interface elements. Floating visual components and organic animations contribute to a clean and contemporary look without compromising system performance.

### 6.2.3 Responsiveness

The system follows a **mobile-first responsive design approach**. CSS grid layouts using auto-fit column structures ensure that the interface adapts smoothly to various screen sizes. This responsiveness is particularly important for scan-and-go features and mobile users accessing the platform.

---

## 6.3 Implementation and Interface Design (Technical Specifics)

The project is developed using a **Modular Full-Stack Architecture**, which separates frontend, backend, and data layers to improve maintainability and scalability.

### 6.3.1 Backend Implementation

The backend is built using **Node.js and Express.js**. It provides RESTful API endpoints for user authentication, medicine management, marketplace transactions, and disposal scheduling. For the prototype phase, a lightweight **JSON-based database (db.json)** is used to ensure fast input/output operations and simplified data handling.

### 6.3.2 Frontend Implementation

The frontend logic is centralized in a controller file (**app.js**) that manages application state, API calls, and dynamic UI rendering. A shared header and footer injection mechanism is used across pages to maintain design consistency and reduce redundancy.

### 6.3.3 Security Mechanisms

Security is enforced using **Role-Based Access Control (RBAC)**, ensuring that each user role such as Buyer, Seller, NGO, and Admin has

controlled access to system functionalities. Additionally, **OTP-based login verification** is implemented to provide an extra layer of protection for sensitive medical transactions.

---

## 6.4 Sell form and disposal center form (for the scheduling )

The image displays two side-by-side web forms against a light green background with a faint leaf pattern. The left form, titled 'Sell Unused Medicine', includes a sub-header 'List your unused medicine for sale. Please ensure the medicine is not expired and is in its original packaging.' and several input fields: 'Medicine Name' (with example 'e.g. AmoxiCare 500mg'), 'Brand / Manufacturer' (with example 'e.g. HealthPharm'), 'Active Ingredient' (with example 'e.g. Amoxicillin'), 'Category' (a dropdown menu showing 'Select Category'), 'Expiry Date' (a date picker showing 'mm/dd/yyyy'), 'Condition' (a dropdown menu showing 'Sealed (Unopened)'), and 'Quantity' (a text input showing '1'). The right form, titled 'Schedule Safe Disposal', includes a sub-header 'Is your medicine expired? Schedule a pickup instead of listing it.' and fields for 'Select Disposal Center' (a dropdown menu showing 'Mumbai Bandra Disposal'), 'Date' (a date picker showing 'mm/dd/yyyy'), 'Time' (a time picker showing '--:--:--'), and 'Notes (Optional)' (a text area with placeholder 'Any special instructions...'). A green 'Schedule Pickup' button is located at the bottom of the right form.

### Unused Medicine Sell Form:

The Unused Medicine Sell Form allows users to submit details of medicines that are unused but still safe for reuse. Through this form, users can provide information such as medicine name, expiry date, quantity, and condition of the medicine. The submitted data helps verified pharmacies review and approve the medicines for redistribution at affordable prices. This form supports the circular economy concept by reducing medicine wastage and improving access to low-cost healthcare resources.

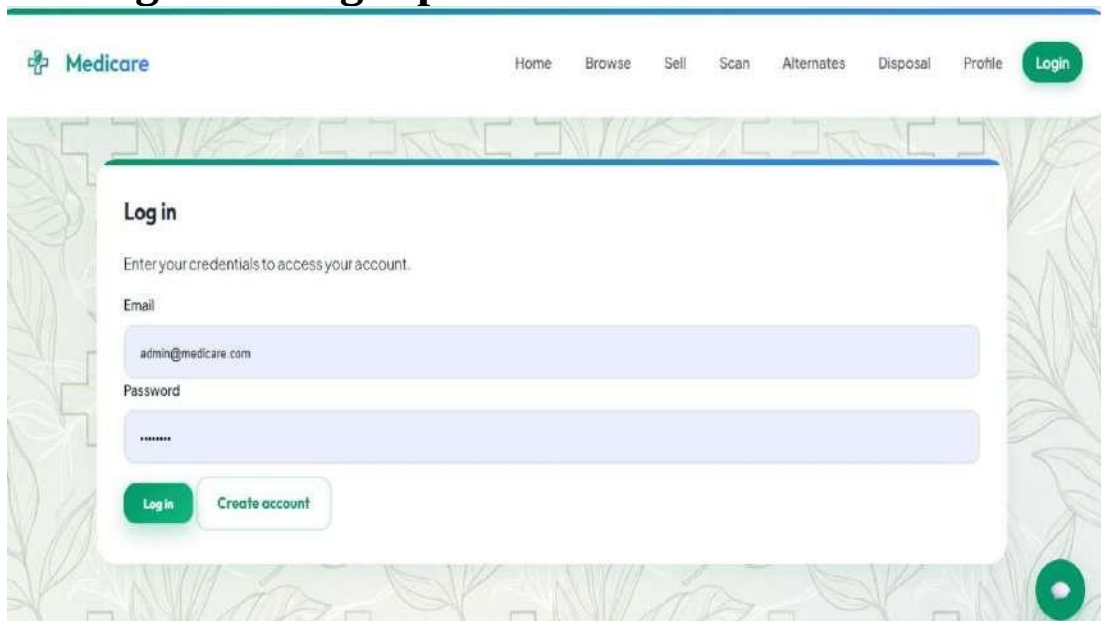
### Disposal Center Form:

The Disposal Center Form is designed to collect information about expired or unsafe medicines that require proper disposal. Users can enter

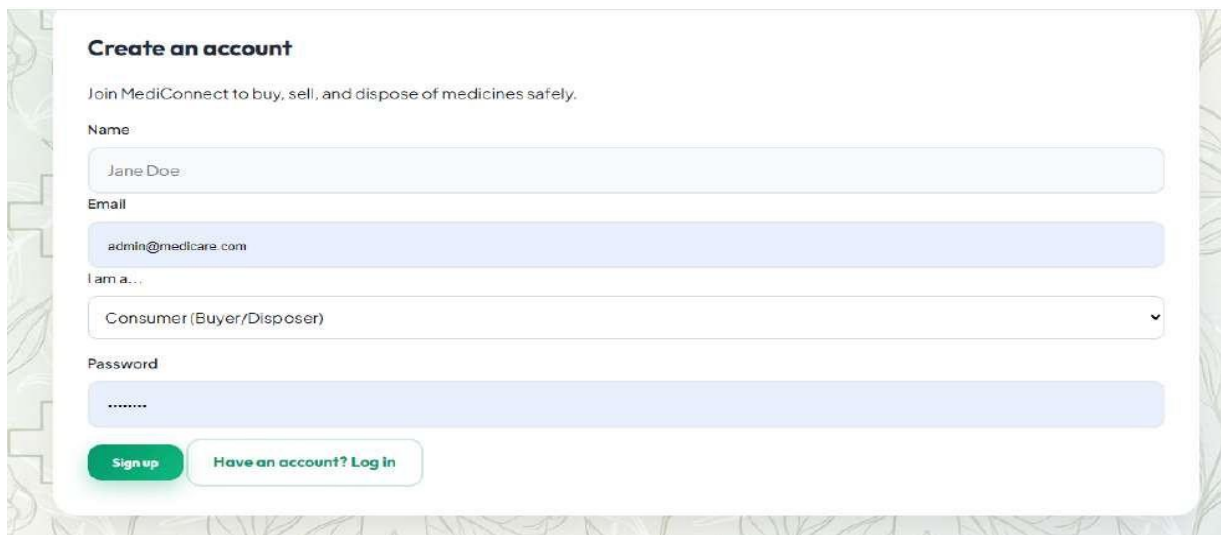
details such as medicine type, quantity, and preferred disposal location. This information is forwarded to authorized disposal or recycling partners to ensure safe and environmentally responsible handling. The form helps prevent improper disposal of medicines, reducing environmental pollution and promoting sustainable healthcare practices.

---

## 6.5 Login and Signup Module



The screenshot shows the Medicare website's login interface. At the top, there is a navigation bar with the Medicare logo and links for Home, Browse, Sell, Scan, Alternates, Disposal, Profile, and a green Login button. The main content area features a large white login box with a green border. Inside the box, the title "Login" is followed by the instruction "Enter your credentials to access your account." Below this are two input fields: "Email" with the text "admin@medicare.com" and "Password" with masked characters. At the bottom of the box are two buttons: a green "Login" button and a white "Create account" button. The background of the page has a subtle pattern of green leaves and pills.



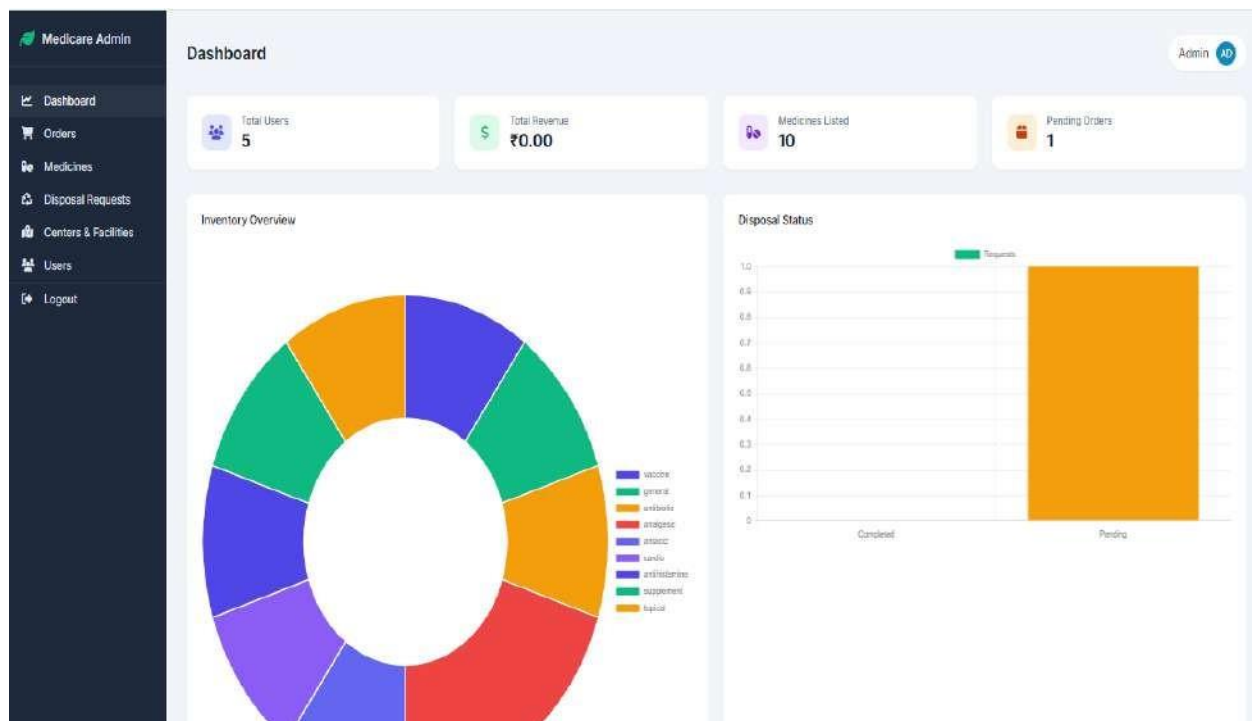
The screenshot shows the Medicare website's "Create an account" page. The title "Create an account" is followed by the instruction "Join MediConnect to buy, sell, and dispose of medicines safely." Below this are four input fields: "Name" with the text "Jane Doe", "Email" with the text "admin@medicare.com", "I am a..." with a dropdown menu showing "Consumer (Buyer/Disposer)", and "Password" with masked characters. At the bottom are two buttons: a green "Sign up" button and a white "Have an account? Log in" button. The background of the page has a subtle pattern of green leaves and pills.

The signup module supports four user roles: Buyer, Seller, NGO, and

Admin. Based on the selected role, the system dynamically configures the user dashboard and accessible features.

The login process follows a **two-step authentication mechanism**. After password verification, a mock One-Time Password (OTP) is generated and displayed in the server console, simulating SMS or email verification. Clean, card-based layouts are used to maintain focus on security.

## 6.6 Dashboard and Marketplace Interface



The dashboard acts as the primary hub for medicine discovery and exchange. Medicines are displayed in a dynamic grid layout showing real-time information such as price, brand, and expiry date.

Users can apply intelligent filters including price-based sorting, expiry-based sorting, and recently added listings. Expiry validation logic continuously checks medicine data, and expired items automatically disable the purchase option and display an **“Expired – Dispose”**

indicator.

## 6.7 Disease Prediction and Alternate Search Module

**Find Alternate Medicines**

Enter a medicine name to find safer, cheaper, or available substitutes with the same composition.

headache Find Substitutes

**Analysis Results**

HEALTH/CONDITION: Headache & Severe Pain  
THERAPEUTIC CLASS: Analgesic / Antipyretic  
ACTIVE INGREDIENT: Propyphenazone + Paracetamol + Caffeine

**Medical Disclaimer:** These suggestions are for informational purposes only. Do not substitute professional medical advice, diagnosis, or treatment. Always consult a doctor before changing medication.

No pharmaceutically equivalent medicines found in current stock.

**Other Market Alternatives (Reference Only)**

|                                                                                                     |              |
|-----------------------------------------------------------------------------------------------------|--------------|
| <b>Saradon</b><br>Class: Analgesic / Antipyretic<br>Active: Propyphenazone + Paracetamol + Caffeine | NOT IN STOCK |
| <b>Dolo 650</b><br>Class: Analgesic / Antipyretic<br>Active: Paracetamol                            | NOT IN STOCK |
| <b>Panadol</b>                                                                                      | NOT IN STOCK |

The disease prediction and alternate search functionality allows users to search for medicines using symptoms or drug names. The system identifies generic compositions and suggests alternative brands with similar effectiveness but varying prices, improving affordability and accessibility

### 6.7 (A) Scanning Module

**Expiry & Info Scanner**

Barcode Label OCR Pill ID

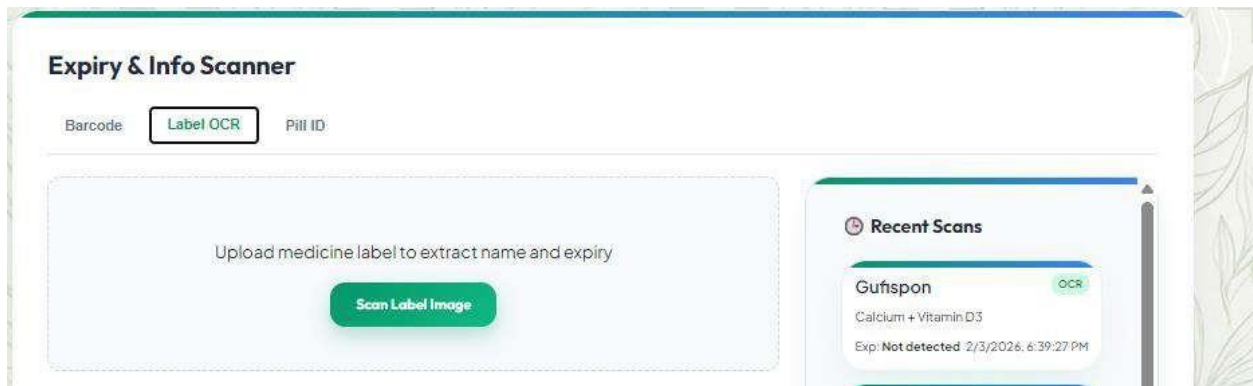
Start Camera Upload Barcode

**Recent Scans**

|                                                                                                        |
|--------------------------------------------------------------------------------------------------------|
| <b>Gufspan</b> <span>OCR</span><br>Calcium + Vitamin D3<br>Exp: Not detected 2/3/2026, 6:39:27 PM      |
| <b>Multivitamin</b> <span>BARCODE</span><br>General Medicine<br>Exp: Not detected 2/3/2026, 6:39:05 PM |

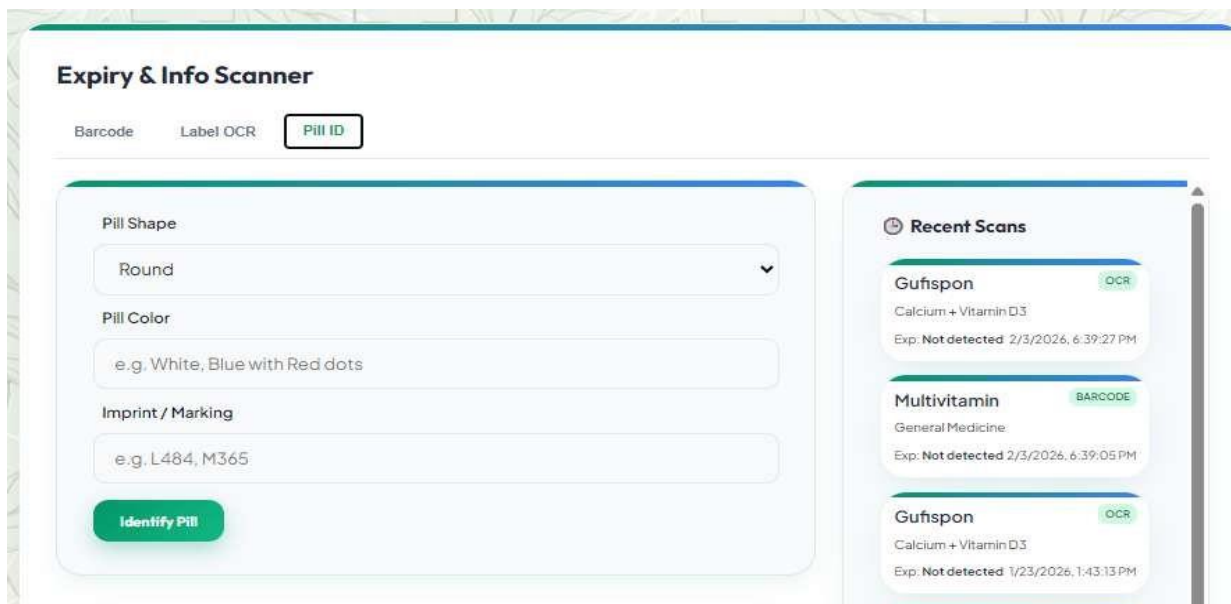
## Barcode Scanner:

The Barcode Search System allows users to scan or upload a medicine barcode to instantly retrieve product details such as medicine name, category, and expiry information. This method is fast, accurate, and reduces manual data entry. It helps users quickly identify medicines and verify whether they are safe for reuse or need disposal.



## Label OCR (Optical Character Recognition):

The Label OCR Search System extracts text information directly from medicine labels using image recognition technology. By scanning the label, the system reads details like medicine name, dosage, and expiry date. This feature is useful when barcodes are missing or damaged and helps automate data collection for unused medicine management.



## Pill ID System:

The Pill ID Search System helps users identify medicines based on physical characteristics such as shape, color, and imprint. Users can upload or scan an image of the pill, and the system matches it with stored data to provide relevant information. This feature improves medicine identification accuracy and enhances safety, especially when original packaging is unavailable.

## 6.8 AI-Based Scoring and Recommendation Engine

**Find Alternate Medicines**

Enter a medicine name to find safer, cheaper, or available substitutes with the same composition.

dolo Find Substitutes

**Analysis Results**

HEALTH CONDITION: Fever & Mild Pain  
THERAPEUTIC CLASS: Analgesic / Antipyretic  
ACTIVE INGREDIENT: Paracetamol

**Medical Disclaimer:** These suggestions are for informational purposes only. Do not substitute professional medical advice, diagnosis, or treatment. Always consult a doctor before changing medication.

☒ Available in inventory

**PainAway Extra**  
Relief Labs  
IBUPROFEN + PARACETAMOL  
₹6.49 Buy Now

[Other Market Alternatives \(Reference Only\)](#)

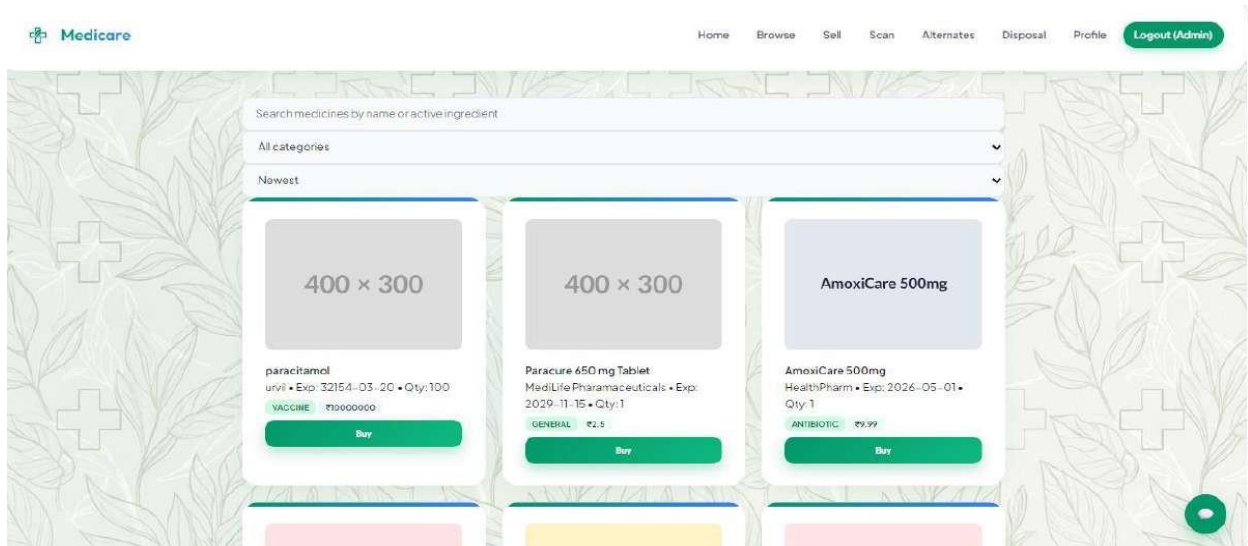
| Other Market Alternatives (Reference Only)                                                          |              |
|-----------------------------------------------------------------------------------------------------|--------------|
| <b>Saradon</b><br>Class: Analgesic / Antipyretic<br>Active: Propyphenazone + Paracetamol + Caffeine | NOT IN STOCK |
| <b>Dolo 650</b><br>Class: Analgesic / Antipyretic<br>Active: Paracetamol                            | NOT IN STOCK |
| <b>Panadol</b><br>Class: Analgesic / Antipyretic<br>Active: Paracetamol                             | NOT IN STOCK |
| <b>Dolo</b><br>Class: Analgesic / Antipyretic<br>Active: Paracetamol                                | NOT IN STOCK |

The platform integrates an **Intent and Scoring Algorithm** to deliver intelligent medicine recommendations. Scores are assigned based on

exact name matches, generic or brand matches, and symptom relevance. Data is aggregated from the local marketplace, a reference medicine dataset, and a mock global knowledge base.

---

## 6.9 Sell and Inventory Listing Interface (Browse)



The Sell and Inventory Listing Interface, implemented in **sell.html**, is designed for verified sellers. Client-side image processing using the FileReader API allows sellers to preview medicine images before submission. Stock management fields such as quantity, batch number, and seller notes ensure transparency. Multi-step form validation prevents invalid price or expiry inputs.

---

## 6.10 Payment Gateway and Secure Checkout

**Secure Checkout**

**Payment Details**

Card Payment ☒ UPI Payment

Cardholder Name  
Medicare customer

Card Number  
4242 4242 4242 4242

Expiry Date  
12/26

CVV  
...

Billing Address  
mumbai

mumbai 401101

**Confirm Payment**

Payments are secure and encrypted.

**Order Summary**

AmoxiCare 500mg AmoxiCare 500mg x 1  
Total: ₹9.99

Subtotal ₹9.99  
Service Fee ₹2.00  
**Total ₹11.99**

**Payment Details**

Card Payment ☐ **UPI Payment** ☒



Scan QR to pay with any UPI App

~ OR ~

Enter UPI ID  
yourname@upi

Billing Address  
mumbai

mumbai 401101

**Confirm Payment**

Payments are secure and encrypted.

**Order Summary**

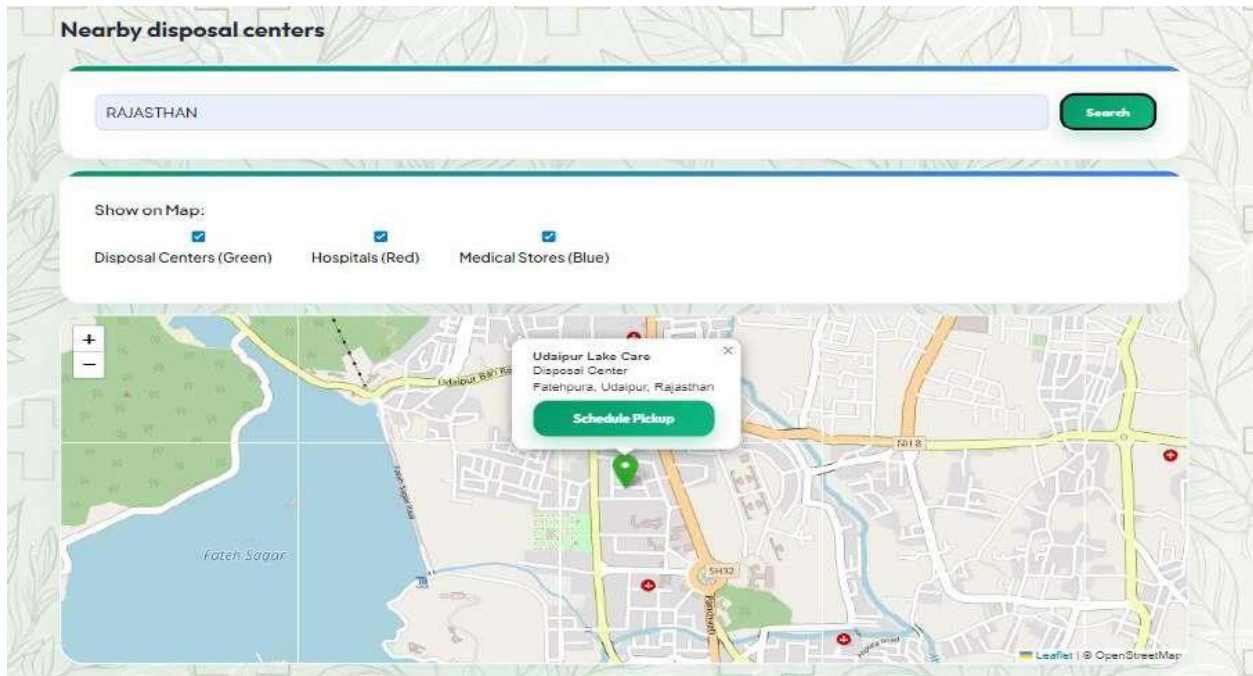
AmoxiCare 500mg AmoxiCare 500mg x 1  
Total: ₹9.99

Subtotal ₹9.99  
Service Fee ₹2.00  
**Total ₹11.99**

The payment module simulates a secure checkout process through **payment.html**. A detailed order summary displays item price, convenience fees, and the final amount. After successful payment, an SMS confirmation is simulated, and inventory quantities are automatically updated to prevent over-selling.

---

## 6.11 Disposal Scheduling and Tracking System



The disposal system allows users to schedule pickups and track disposal status through stages such as Pending, In Transit, and Safely Disposed. Pickup notes and time-slot selection improve coordination with NGO partners, while disposal history promotes user accountability.

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## 6.12 Admin Command Center

The Admin Command Center provides administrators with a centralized dashboard to monitor system metrics such as total users, listed medicines, and completed disposals. Admins can oversee all orders and moderate inventory to remove unsafe or suspicious listings.

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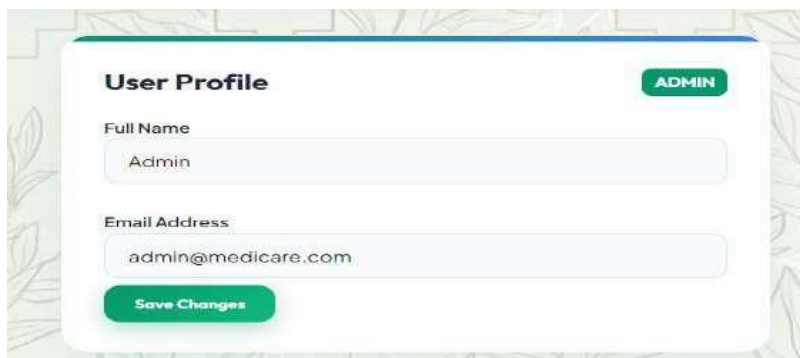
## 6.13 Healthcare Education Module



The healthcare education module focuses on spreading awareness about safe medicine disposal. An accordion-style interface explains the risks of improper disposal and guides users toward environmentally responsible practices.

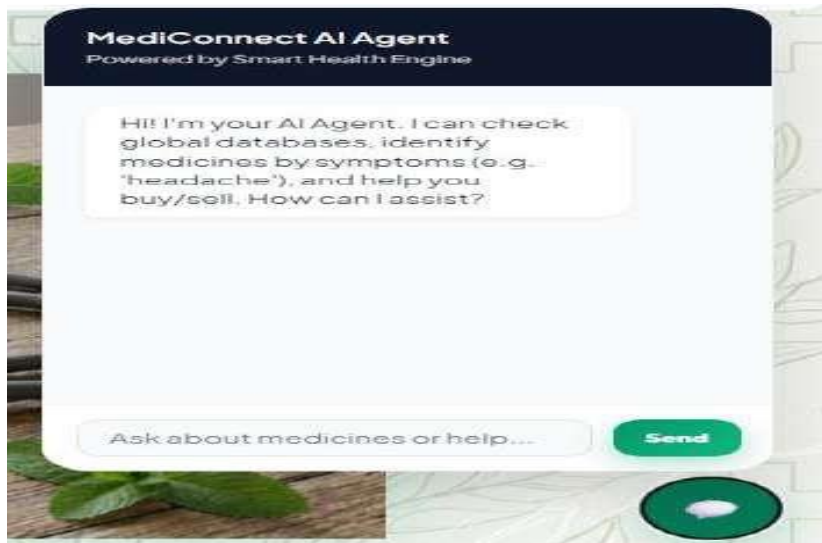
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## 6.14 Profile and Role Management



The profile management module allows users to view personal details such as join date, role badge, and account ID. Contextual quick links are provided based on user roles, enhancing usability and navigation efficiency.

## 6.15 AI CHATBOT



My chatbot is an AI-based conversational assistant integrated into the Medicare platform to support users in managing unused and expired medicines responsibly. It helps households by answering queries related to medicine reuse, safe disposal methods, nearby verified pharmacies, and recycler or disposal partners. Using natural language processing (NLP), the chatbot understands user questions and provides instant guidance, improving awareness, accessibility, and decision-making. By automating support and recommendations, the chatbot enhances user experience while promoting sustainability, affordability, and environmental safety in the healthcare ecosystem.

# Chapter 7 : Conclusion

## 7.1 Conclusion

The development of the **Medicare platform** successfully demonstrates how modern information technology can act as a powerful enabler of a **circular economy in the healthcare domain**. The system effectively bridges the gap between individual households, verified pharmacies, and authorized disposal partners by providing a unified digital platform for the complete lifecycle management of medicines.

Unlike traditional healthcare models that treat unused or expired medicines as waste, Medicare introduces a **sustainable and responsible ecosystem** where safe, unused medicines can be redistributed and expired medicines can be disposed of correctly. This approach not only reduces unnecessary medical waste but also promotes **economic affordability, environmental safety, and social responsibility**.

By integrating intelligent features such as an AI-powered Health Agent, barcode-based expiry validation, and map-based disposal center discovery, the system simplifies complex healthcare decisions for common users. Overall, Medicare proves that technology-driven solutions can significantly enhance healthcare accessibility while maintaining safety, transparency, and sustainability.

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### 7.1.1 Significance of the System

The Medicare system holds significant value due to its multi-dimensional impact on society, the environment, and technology. The

major areas of significance are outlined below:

## 1. Societal Impact

Medicare improves healthcare accessibility for economically weaker sections by enabling access to **verified, unused medicines at reduced costs** through its digital marketplace. This reduces financial barriers for low-income patients and supports inclusive healthcare delivery.

## 2. Environmental Impact

Improper disposal of medicines often leads to contamination of soil and water bodies. Medicare addresses this issue by offering an **interactive map-based disposal system** along with user education, ensuring that expired or unusable medicines are disposed of responsibly and safely.

## 3. Technological Impact

The integration of an **AI-driven Health Agent** and **Barcode Expiry Scanner** helps users make informed decisions regarding medicine usage. These features reduce the risk of accidental consumption of expired drugs and minimize dependency on manual verification, making healthcare technology more user-friendly and reliable.

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## 7.2 Limitations of the System

Although the Medicare system is fully functional as a prototype, certain limitations exist that need to be addressed before large-scale deployment:

### 1. Clinical Verification

The platform currently depends on seller-provided information and a static reference dataset. It does not yet support real-time **Doctor's Prescription Verification**, especially for regulated medicines such as

Schedule-H drugs.

## 2. Mock Dependencies

Certain functionalities, including the **Barcode Scanner** and **OTP verification gateway**, operate using simulated logic for demonstration purposes. These features will require integration with live APIs to function in real-world environments.

## 3. Data Storage Constraints

At present, the system uses a **JSON-based flat file (db.json)** for data storage. While this is suitable for prototyping, it is not ideal for high-concurrency environments. A scalable database such as **PostgreSQL** or **MongoDB** would be necessary for production-level deployment.

## 4. Geographic Coverage

The disposal center mapping is currently optimized for major urban regions such as Mumbai and Delhi. To ensure nationwide usability, additional data ingestion is required to cover **semi-urban and rural areas**.

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## 7.3 Future Scope of the Project

The future scope of Medicare focuses on transforming the prototype into a **robust, production-ready healthcare utility**. Several enhancements can be incorporated to extend its impact and scalability:

### 1. Cloud Integration

Deploying the backend infrastructure on cloud platforms such as **AWS (Amazon Web Services)** or **Google Cloud** would ensure high availability, better performance, scalability, and approximately **99.9% uptime**.

## 2. Blockchain-Based Transparency

Implementing **blockchain technology** can enable secure tracking of medicine provenance. This would allow every listed medicine to be traced from its original purchase to its current availability, improving trust, transparency, and fraud prevention.

## 3. IoT-Enabled Disposal Bins

Smart disposal bins integrated with **IoT sensors** can automatically notify the system when they are full and ready for pickup. This automation would enhance operational efficiency and improve waste management logistics.

## 4. Machine Learning Enhancements

The AI Health Agent can be upgraded with advanced **Natural Language Processing (NLP)** techniques to analyze complex symptom descriptions and improve disease-matching accuracy and recommendation quality.

## 5. Mobile Native Application

Developing dedicated **Android and iOS applications** using **React Native** would allow the system to leverage native device features such as low-light barcode scanning, push notifications, and real-time expiry alerts, thereby improving user engagement and accessibility.

## References

1. **Node.js Official Documentation.**  
Available at: <https://nodejs.org/en/docs/>
2. **Express.js Documentation.**  
Available at: <https://expressjs.com/>
3. **React Official Documentation.**  
Available at: <https://react.dev/>
4. **MongoDB Documentation.**  
Available at: <https://www.mongodb.com/docs/>
5. **Mongoose Documentation.**  
Available at: <https://mongoosejs.com/docs/>
6. **JSON Web Token (JWT) Documentation.**  
Available at: <https://jwt.io/>
7. **Bootstrap Documentation.**  
Available at: <https://getbootstrap.com/docs/>
8. **HTML5 – W3C Specification.**  
Available at: <https://www.w3.org/TR/html52/>
9. **CSS3 – W3C Specification.**  
Available at: <https://www.w3.org/Style/CSS/>
10. **JavaScript (ECMAScript) Documentation – MDN Web Docs.**  
Available at: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>